

E-COMMERCE SALES ANALYSIS REPORT

Sales Performance | Revenue Insights | Product Analysis

Dataset: 1,000 Products | 7 Categories | 12 Months of Sales Data

Tools Used: Python, PostgreSQL, Power BI

Executive Summary

This report presents a comprehensive analysis of e-commerce sales performance using transactional product-level data containing 1,000 products distributed across seven categories. The primary objective of the analysis was to identify revenue-driving categories, product performance trends, seasonal sales behavior, pricing patterns, and customer review impacts on business outcomes.

The analysis was performed using PostgreSQL for data extraction and business intelligence querying, Python for exploratory data analysis, and Power BI for dashboard development and visualization.

Key Performance Indicators

KPI	Value
Total Products	1,000
Total Units Sold	6,019,912
Total Revenue	₹1.49 Billion
Categories	7
Avg. Revenue per Product	₹1.49 Million
Best Performing Category	Books

Key Findings

- Books emerged as the highest revenue-generating category.
- October recorded the strongest sales performance.
- Revenue is concentrated among a relatively small number of top-performing products.
- Customer ratings exhibit minimal correlation with sales performance.
- Home & Kitchen generated the lowest revenue contribution among all categories.

These findings provide actionable insights that can improve inventory planning, category management, marketing effectiveness, and long-term revenue growth.

1. Business Objectives

The analysis was designed to address the following business questions:

Revenue Analysis

- Which categories generate the highest revenue?
- What products contribute most to overall revenue?
- How concentrated is revenue among top products?

Sales Trend Analysis

- Which months contribute the highest sales?
- Are there identifiable seasonal patterns?

Product Performance Analysis

- Which products consistently outperform others?
- Which products require intervention or discontinuation?

Customer Review Analysis

- Do higher ratings lead to higher sales?
- Which categories receive the best customer feedback?

Strategic Planning

- How can inventory planning be optimized?
- Which categories should receive greater investment?

2. Dataset Overview & Methodology

Dataset Description

The dataset contains sales and product information for 1,000 products.

Variables

Variable	Type	Description
Product ID	Integer	Unique product identifier
Product Name	Text	Product description
Category	Text	Product category

Variable	Type	Description
Price	Decimal	Product selling price
Review Score	Float	Customer rating
Review Count	Integer	Number of customer reviews
Sales Month 1-12	Integer	Monthly units sold

Data Preparation

The following preprocessing steps were performed:

Data Validation

- Verified data integrity
- Checked for duplicate records
- Confirmed data type consistency

Feature Engineering

Created:

Annual Sales

Annual Sales = Sum(Month1 ... Month12)

Annual Revenue

Annual Revenue = Annual Sales × Price

These metrics formed the foundation of all revenue-based analyses.

3. Revenue Analysis

3.1 Overall Revenue Performance

The platform generated approximately:

₹1.49 Billion Revenue

with over:

6 Million Units Sold

This indicates strong business performance and significant customer demand across categories.

3.2 Revenue by Category

Category	Revenue (₹ Million)
Books	236.8
Sports	232.6
Toys	230.2
Health	221.7
Electronics	201.7
Clothing	187.3
Home & Kitchen	178.4

Insight

Books contribute the highest revenue despite not necessarily being the highest-priced category.

Sports and Toys also perform strongly, indicating high consumer demand.

Business Implication

Management should prioritize:

- Inventory investment
- Marketing campaigns
- Product expansion

within these categories.

3.3 Revenue Share Analysis

Top 3 Categories contribute more than 45% of total revenue.

This suggests revenue concentration among specific consumer interests.

Recommendation

Expand successful product lines in Books and Sports while optimizing weaker categories.

4. Sales Trend Analysis

Monthly Sales Performance

Month	Units Sold
January	498,122
February	507,661
March	506,739
April	498,775
May	500,482
June	499,321
July	507,011
August	500,991
September	502,374
October	514,798
November	505,838
December	478,800

Peak Sales Month

October

Generated the highest sales volume.

Possible reasons:

- Festive season demand
- Promotional campaigns
- Increased consumer spending

Lowest Sales Month

December

Recorded relatively weaker sales compared to October.

Recommendation

Run targeted end-of-year promotional campaigns to improve performance.

5. Product Performance Analysis

Top Revenue Generating Products

Product	Revenue
Product 306	₹3.81 Million
Product 532	₹3.76 Million
Product 53	₹3.72 Million
Product 229	₹3.70 Million
Product 391	₹3.65 Million

Key Insight

A small number of products contribute disproportionately to total revenue.

This follows the Pareto Principle commonly observed in retail environments.

Recommendation

- Prioritize stock availability
 - Feature these products prominently
 - Use them in promotional campaigns
-

Bottom Performing Products

Products with low annual sales should be reviewed for:

- Pricing issues
- Product relevance
- Marketing effectiveness

Potential discontinuation candidates should be evaluated.

6. Category Deep Dive

Books

Strengths

- Highest revenue category
- Consistent monthly demand

Opportunities

- Expand catalog
 - Bundle recommendations
-

Sports

Strengths

- Strong demand
- High customer engagement

Opportunities

- Seasonal campaigns
 - Cross-selling accessories
-

Toys

Strengths

- High volume category

Opportunities

- Holiday promotions
 - Gift bundles
-

Electronics

Strengths

- Premium pricing potential

Opportunities

- Upselling warranties
 - Accessory bundles
-

Health

Strengths

- Stable demand

Opportunities

- Subscription models
 - Repeat purchase incentives
-

Clothing

Strengths

- Broad customer appeal

Opportunities

- Personalized recommendations
-

Home & Kitchen

Challenges

- Lowest revenue category

Recommendation

- Reevaluate assortment
 - Improve category visibility
-

7. Pricing Analysis

Product Price Distribution

Understanding pricing patterns helps determine whether revenue is driven by premium products or high-volume low-price products.

Observations

- Mid-priced products dominate sales volume.
- Premium products contribute significantly to revenue.
- Extremely low-priced products contribute less to overall revenue.

Business Implication

Balanced pricing strategies appear most effective.

8. Customer Review Analysis

Average Review Score

Customer ratings remain relatively consistent across categories.

Review Score vs Sales

Correlation:

-0.018

This indicates almost no relationship between ratings and sales.

Review Count vs Sales

Correlation:

-0.069

Again, a very weak relationship.

Insight

Sales appear to be driven primarily by:

- Category demand
- Pricing
- Product visibility

rather than customer ratings.

9. SQL Analysis

The project utilized advanced PostgreSQL concepts.

Aggregate Functions

Used:

- SUM()
- AVG()
- MIN()
- MAX()

Purpose:

Revenue and sales analysis.

Ranking Functions

Used:

- RANK()
- DENSE_RANK()
- ROW_NUMBER()

Purpose:

Product and category rankings.

Window Functions

Used:

- Running totals
- Revenue rankings
- Cumulative sales

Purpose:

Trend and contribution analysis.

Common Table Expressions (CTEs)

Used to simplify:

- Revenue calculations
 - Category summaries
 - Product rankings
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10. Power BI Dashboard Analysis

An interactive dashboard was created to support stakeholder decision-making.

Dashboard Components

Executive Overview

KPIs:

- Revenue
 - Sales
 - Products
 - Categories
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Category Dashboard

Visuals:

- Revenue by Category
- Revenue Share
- Category Ranking

Product Dashboard

Visuals:

- Top Products
 - Revenue Contribution
 - Product Ranking
-

Trend Dashboard

Visuals:

- Monthly Sales Trend
 - Seasonal Analysis
-

Review Dashboard

Visuals:

- Rating Distribution
 - Review Count Analysis
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11. Key Findings

Finding 1

Books generate the highest revenue.

Finding 2

October is the strongest sales month.

Finding 3

Revenue concentration exists among a small group of products.

Finding 4

Customer ratings have negligible influence on sales.

Finding 5

Home & Kitchen underperforms compared to other categories.

12. Strategic Recommendations

Revenue Growth

Increase investment in:

- Books
 - Sports
 - Toys
-

Inventory Optimization

Prepare inventory ahead of:

- October
- November

to avoid stock shortages.

Product Strategy

Expand:

- Top-selling product lines

Review:

- Low-performing products
-

Marketing Strategy

Implement:

- Cross-selling
- Product recommendations
- Category-focused campaigns

Data Strategy

Collect:

- Customer demographics
- Repeat purchase behavior
- Marketing attribution data

for future predictive analytics initiatives.

13. Conclusion

The E-Commerce Sales Analysis successfully identified revenue drivers, seasonal patterns, category-level opportunities, and product performance trends. The findings demonstrate the value of data analytics in supporting strategic business decisions and improving operational efficiency.

The integration of PostgreSQL, Python, and Power BI enabled comprehensive analysis and actionable recommendations. Future enhancements should include customer-level behavioral data, profitability metrics, and predictive forecasting models to further strengthen business intelligence capabilities.

Appendix

Technologies Used

- PostgreSQL
- Python
- Pandas
- NumPy
- Power BI

Analytical Techniques

- Exploratory Data Analysis
- Revenue Analysis
- Product Ranking
- Category Segmentation
- Correlation Analysis
- Dashboard Development

Deliverables

- SQL Analysis Scripts
- Power BI Dashboard
- Data Analytics Report
- Business Recommendations

— End of Report —